Chapter 6: Pointers and Memory Management

Introduction

Pointers are among the most powerful features in languages like C and C++. They offer direct access to memory, enabling dynamic memory management, efficient data handling, and low-level system programming. However, with great power comes great responsibility: improper use of pointers can lead to bugs, memory leaks, and program crashes. This chapter will explore pointers, dynamic memory allocation, and pointer arithmetic with arrays in detail.

Understanding Pointers

A **pointer** is a variable that stores the memory address of another variable. Instead of storing a value directly, it "points" to the location where the value is stored. This capability makes pointers fundamental for tasks that require dynamic memory allocation, array handling, and function arguments passing by reference.

Declaration and Initialization of Pointers

A pointer is declared using the `*` operator. Here's the syntax:

```
int *ptr;  // Declaration of a pointer to an integer
```